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Parameter-efficient prompt tuning based on adaptive fused pre-trained models for 3D scene understanding

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ABSTRACT

Prompt tuning in point cloud pre-trained models has achieved significant reductions in trainable parameters while maintaining promising performance, showing remarkable potential in 3D vision tasks. However, the existing methods of generating prompts from a single pre-trained model may suffer from limited expressive capability, which potentially leads to performance degradation when encountering noise, outliers, or adversarial samples. To alleviate this limitation, we propose a novel parameter-efficient prompt tuning (PEPT) framework, based on adaptive fused pre-trained models. Our framework leverages prompt information generated by diverse pre-trained models to capture various point cloud features, enhancing the model's ability to learn more comprehensive representations. Specifically, we introduce a multi-level prompt generation (MLPG) module for pre-trained point cloud models, which can efficiently aggregate features at different levels. By utilizing richer and more diverse feature representations, it generates more discriminative prompts while significantly reducing the trainable parameters of the module. Furthermore, we designed a lightweight task head (LightHead). The features extracted from each pre-trained model are adaptively fused and fed into the LightHead. By jointly optimizing different models, the robustness of the framework is improved. Extensive experiments demonstrate that our PEPT framework achieves competitive performance while using only 2% of the trainable parameters compared to full fine-tuning on most tasks.

1. Introduction

Point clouds, which consist of sets of three-dimensional (3D) coordinates, are often accompanied by additional features such as color or intensity (Baicocchi et al., 2025; Chen et al., 2026; Sun et al., 2025; Yuan et al., 2025; Zhang et al., 2025b). As a fundamental 3D representation, point clouds provide rich, precise geometric and spatial information for 3D scene understanding (Asad et al., 2026; Li et al., 2025c,d; Shi et al., 2025; Yoon et al., 2026), and help computers accurately perceive and interpret spatial structures, object shapes, and their interrelationships in the real world. Point clouds have been widely studied and adopted in robotics perception (Cheng et al., 2022), autonomous driving (Huang et al., 2023b; Jati et al., 2025; Lu et al., 2024; Zhang et al., 2021), and virtual/augmented reality (Blanc et al., 2020; Genay et al., 2021). 3D point cloud understanding (Chen et al., 2026; Hu et al., 2025; Song et al., 2026; Sun et al., 2026; Zhang et al., 2025a) aims to extract meaning-

ful semantic and geometric information from unstructured point cloud data, enabling machines to perceive, interpret, and reason about complex 3D environments, which is very important for the application scenarios above. However, many existing 3D point cloud understanding methods are supervised and require high-quality annotated data, which is costly and labor-intensive.

To alleviate the reliance on annotated data, self-supervised learning (Li et al., 2023b, 2022b; Liu et al., 2022a; Xu et al., 2025) has emerged as an effective alternative. Recent approaches such as Point-BERT (Yu et al., 2022), Point-MAE (Pang et al., 2022), ACT (Dong et al., 2023), RECON (Qi et al., 2023), and Point-GPT (Chen et al., 2023) leverage large-scale unlabeled data to learn transferable representations, enhancing the performance of models in downstream tasks. However, these methods often require substantial storage and computational cost for fine-tuning, limiting their deployment in resource-constrained environments.

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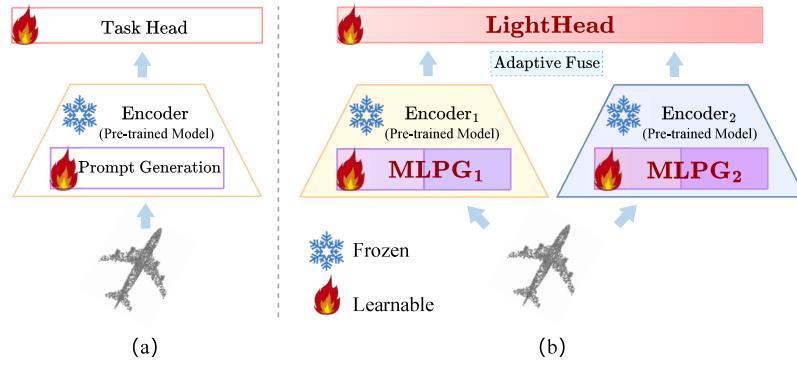


Fig. 1. (a) Conventional prompt generation and task head from a single pre-trained model. (b) Our proposed PEPT with the multi-level prompt generation (MLPG) module and the lightweight task head (LightHead), which is based on different pre-trained models.

Parameter-efficient fine-tuning (PEFT) (Han et al., 2024), especially prompt tuning, addresses this limitation by optimizing only a small set of parameters while keeping the pre-trained model parameters frozen. Recent studies, including IDPT (Zha et al., 2023), DAPT (Zhou et al., 2024), and ADAPTER-X (Li et al., 2024), have demonstrated that prompt tuning can achieve comparable performance to full fine-tuning with significantly fewer trainable parameters in point cloud analysis. Although these studies can optimize point cloud models in a more efficient manner, they are based on single pre-trained models, resulting in limited expressive capability, rendering them susceptible to noise, outliers, and adversarial samples. This raises concerns in safety-critical applications (e.g., autonomous driving and medical diagnosis), where robust and reliable performance is essential.

To this end, in this work, we consider integrating features from diverse pre-trained models that have distinct prompt information. This strategy mitigates individual model biases while generating richer, more expressive feature representations. For example, the pre-trained model Point-BERT (Yu et al., 2022) can capture global semantic information, while Point-MAE (Pang et al., 2022) emphasizes local geometric features. To be specific, Point-BERT (Yu et al., 2022), based on a semantic masking pre-training task, uses a multi-head attention mechanism focusing more on modeling global dependencies in point cloud data and capturing global semantic information. Point-MAE (Pang et al., 2022), based on an autoencoding pre-training task, employs a more compact encoder-decoder architecture to emphasize local geometric feature modeling.

Based on this idea, we propose a novel parameter-efficient prompt tuning (PEPT) framework, based on adaptive fused pre-trained models. Fig. 1 shows the comparison between conventional prompt generation and task head from a single pre-trained model and our multi-level prompt generation (MLPG) and lightweight task head (LightHead) based on different pre-trained models. PEPT integrates features from multiple pre-trained models that have their own prompt information. In this way, PEPT can exploit complementary strengths from diverse pre-trained models, generating richer and more comprehensive feature representations, reducing biases inherent in individual pre-trained models.

Considering that directly combining multiple pre-trained models increases the trainable parameters, we introduce an MLPG module that decreases the parameters for generating prompts. In addition, we develop a LightHead, which adaptively fuses each pre-trained model's features based on input and task requirements, enhancing the model's robustness and adaptability in downstream tasks.

Extensive experiments on the ModelNet40 (Wu et al., 2015), ScanObjectNN (Uy et al., 2019), and ShapeNetPart (Yi et al., 2016) datasets demonstrate the effectiveness of PEPT in point cloud classification, few-shot learning, and part segmentation tasks. Our method achieves competitive performance while fine-tuning fewer parameters.

Our main contributions are summarized as follows:

- We propose a novel parameter-efficient prompt tuning (PEPT) framework, based on adaptive fused pre-trained models. It integrates diverse models and leverages their complementary strengths.
- We introduce a multi-level prompt generation (MLPG) module that efficiently generates prompts for the co-representation of features at different levels. The number of trainable parameters is reduced while learning more discriminative prompt information.
- We present a lightweight task head (LightHead) that effectively learns and combines information from multiple pre-trained models. Optimize each model according to the input features and task requirements.
- Extensive experiments on challenging benchmark datasets, including ModelNet40, ScanObjectNN, and ShapeNetPart, demonstrate that PEPT achieves competitive performance with only 0.4M trainable parameters in most tasks.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on pre-training point cloud models, point cloud prompt tuning, and multi-model learning methods. Section 3 details our PEPT framework and its theoretical foundations. Section 4 presents comprehensive experimental results and analyses, and the paper concludes in Section 5.

2. Related work

2.1. Pre-training point cloud models

Point cloud (Feng et al., 2020; Gao et al., 2025; He et al., 2025; Huang et al., 2026; Ryu et al., 2025) provides highly accurate and detailed 3D representations of physical spaces and objects for 3D scene understanding (Ai et al., 2025; Bai et al., 2024; Du et al., 2025; Ma et al., 2025; Yu et al., 2025). Building on the success of deep learning (Jiao et al., 2025a, 2023, 2025b, 2022) in natural image understanding, researchers are now extending these techniques to point clouds, aiming to unlock the rich spatial information embedded in 3D data. Given the scarcity of labeled data in natural images (Du et al., 2022; Hao et al., 2025; Li et al., 2023a, 2025a), obtaining labeled point clouds is even more difficult due to the complexity of 3D data. Recent advances in 3D vision have shifted toward self-supervised learning (Yang et al., 2022) to address the scarcity of annotated data. Abundant unlabeled point cloud data is used to learn feature representations and reduce dependence on annotations. PointContrast (Xie et al., 2020) and CrossPoint (Afham et al., 2022) adopt self-supervised contrastive learning to enhance the model's ability to recognize 3D objects. Point-BERT (Yu et al., 2022), Point-MAE (Pang et al., 2022), and Point-M2AE (Zhang et al., 2022) use masked modeling to learn feature representations of point clouds. ACT (Dong et al., 2023) utilizes pre-trained 2D images or natural language

transformers for self-supervised 3D representation learning. PointGPT (Chen et al., 2023) introduces the concept of GPT to point clouds and presents a point cloud auto-regressive generation task to pre-train transformer models. RECON (Qi et al., 2023) combines the strengths of generative modeling and contrastive learning to improve 3D representation learning. These methods have demonstrated impressive performance in various downstream tasks, including classification (Uy et al., 2019; Wu et al., 2015) and segmentation (Yi et al., 2016).

Despite their effectiveness in learning rich 3D representations, these pre-training methods typically require full fine-tuning of the entire backbone for downstream task adaptation. Full fine-tuning not only demands substantial computational resources but also risks altering the well-learned representations in pre-trained models.

2.2. Prompt tuning methods

The growing demand for PEFT (Wang et al., 2025a) in 3D point cloud processing has spurred research into methods that achieve high performance with minimal trainable parameters. Prompt tuning (Jia et al., 2022; Li et al., 2025b; Liu et al., 2025; Wang et al., 2024a,b) enables adaptation to different downstream tasks by fine-tuning only a small set of parameters rather than the entire pre-trained model. Sohn et al. (2023) pioneered visual prompt tuning for generative transfer learning, demonstrating how learnable prompts enable parameter-efficient domain adaptation of frozen vision transformers. UniSOD (Wang et al., 2025b) proposes a unified framework for single-modal and multi-modal salient object detection via adaptive prompt learning. Recent works have demonstrated the effectiveness of prompt tuning in point cloud analysis. IDPT (Zha et al., 2023) proposes a dynamic prompt generation module that captures instance-specific semantic features from point clouds. ADAPTER-X (Li et al., 2024) introduces a general PEFT framework for 2D images and 3D point clouds. Point-PEFT (Tang et al., 2024) develops a novel approach that combines a point-prior prompt with a geometry-aware adapter, efficiently incorporating downstream 3D semantics into the pre-trained models. DAPT (Zhou et al., 2024) learns the representation features through the combination of a dynamic adapter and prompt tuning mechanisms. These approaches achieve comparable or superior performance to full fine-tuning while requiring fewer trainable parameters.

However, existing prompt tuning methods (Li et al., 2024; Sohn et al., 2023; Tang et al., 2024; Wang et al., 2025b; Zha et al., 2023; Zhou et al., 2024) predominantly rely on single pre-trained models, making them susceptible to noise, outliers, and adversarial samples. This limitation poses significant challenges for safety-critical applications like autonomous driving and medical diagnosis, where robust and reliable performance is paramount.

2.3. Multi-model learning methods

Inspired by the success of transformers in natural language processing (Devlin et al., 2019; Vaswani et al., 2017) and computer vision (Dosovitskiy et al., 2021; Huang et al., 2023a; Li et al., 2022a; Liu et al., 2021), some studies have introduced transformers into multi-model frameworks and demonstrated significant advantages. CrossViT (Chen et al., 2021) uses two separate transformer branches of different computational complexities to learn multi-scale feature representations. MPViT (Lee et al., 2022) uses multiple parallel transformers to process image features at different scales, obtaining both fine and coarse feature representations. These studies demonstrate that multi-modal approaches yield more robust and effective feature representations. TSC-Mamba (Ahmed & Cheng, 2025) obtains more robust features by fusing multi-view time-series representations, which shares a similar motivation with our adaptive fusion strategy for multiple pre-trained models.

Although multi-model approaches have achieved success in computer vision, their potential in point cloud prompt learning remains

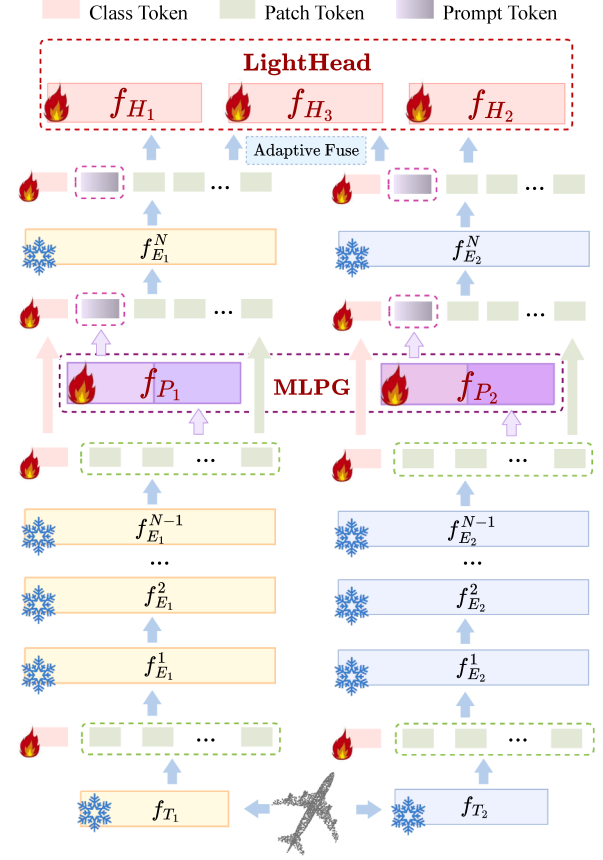


Fig. 2. The overview of the proposed PEPT. Two different encoders produce outputs from their second-to-last layers ($f_{E_1}^{N-1}$ and $f_{E_2}^{N-1}$), which are fed into separate MLPG modules (f_{P_1} and f_{P_2}) to generate multi-level prompt tokens. The features extracted from the two branches are fed into the LightHead (f_{H_1} , f_{H_2} , and f_{H_3}), and the final output is obtained by f_{H_3} .

largely unexplored. To leverage these benefits in the point cloud understanding task, we propose integrating features from multiple models that have distinct prompt information.

3. Methodology

In this section, we delineate our PEPT designed for 3D scene understanding. We first introduce the overall framework of PEPT in Section 3.1 and then present the architecture of the MLPG module in Section 3.2, which efficiently generates prompts using different level features. Finally, we describe our LightHead to learn and adaptively combine information from diverse pre-trained models in Section 3.3.

3.1. Overview

Current point cloud prompt tuning methods typically rely on single pre-trained models, making them vulnerable to noise, outliers, and adversarial samples. To enhance the robustness of the network, we introduce a novel PEPT that efficiently generates and integrates information from diverse pre-trained point cloud models.

Without loss of generality, let's take two branches as examples, denoted f_{E_1} and f_{E_2} , respectively. As shown in Fig. 2, PEPT is composed of two point cloud feature extraction branches and a lightweight task head. Insert MLPG modules f_{P_1} and f_{P_2} on branches f_{P_1} and f_{P_2} , respectively. The downstream task head (LightHead) is composed of f_{H_1} , f_{H_2} , and f_{H_3} for classification or segmentation. f_{H_1} and f_{H_2} are the

task heads of the two pre-training models, respectively. f_{H_3} processes the fused features of the two branches to produce the final results.

Specifically, for each pre-trained branch, an input point cloud is partitioned into local patches, embedding patches into tokens with positional encoding via encoders f_{T_1} and f_{T_2} , followed by inserting classification tokens at the head of the patch tokens, and processed through the N encoding layers. The classification tokens $c_{E_1}^{N-1}$, $c_{E_2}^{N-1}$ and patch tokens $T_{E_1}^{N-1}$, $T_{E_2}^{N-1}$ are extracted from the $(N-1)$ th transformer encoding layer. These tokens are fed into the corresponding prompt generation modules f_{P_1} and f_{P_2} to produce different prompt tokens $p_{E_1}^{N-1}$, $p_{E_2}^{N-1}$:

$$p_{E_1}^{N-1} = f_{P_1}[c_{E_1}^{N-1}; T_{E_1}^{N-1}], \quad (1)$$

$$p_{E_2}^{N-1} = f_{P_2}[c_{E_2}^{N-1}; T_{E_2}^{N-1}]. \quad (2)$$

The classification tokens $c_{E_1}^N$, $c_{E_2}^N$, prompt tokens $p_{E_1}^N$, $p_{E_2}^N$, and patch tokens $T_{E_1}^N$, $T_{E_2}^N$ in the N th encoding layer are described as follows:

$$[c_{E_1}^N; p_{E_1}^N; T_{E_1}^N] = f_{E_1}^N[c_{E_1}^{N-1}; p_{E_1}^{N-1}; T_{E_1}^{N-1}], \quad (3)$$

$$[c_{E_2}^N; p_{E_2}^N; T_{E_2}^N] = f_{E_2}^N[c_{E_2}^{N-1}; p_{E_2}^{N-1}; T_{E_2}^{N-1}]. \quad (4)$$

Let g_1 and g_2 denote the task heads of the two pre-trained models. To achieve a more discriminative feature representation, we employ a fusion strategy for the features derived from the two branches. The unified task head g_3 combines features from both branches to produce the final output:

$$g_1 = f_{T_1}[c_{E_1}^N; p_{E_1}^N; T_{E_1}^N], \quad (5)$$

$$g_2 = f_{T_2}[c_{E_2}^N; p_{E_2}^N; T_{E_2}^N], \quad (6)$$

$$g_3 = f_{T_3}[c_{E_1}^N; p_{E_1}^N; T_{E_1}^N; c_{E_2}^N; p_{E_2}^N; T_{E_2}^N]. \quad (7)$$

Through these steps, PEPT introduces learnable prompt tokens into multiple models to capture information from different models, thereby providing richer and more effective guidance for the downstream tasks. Our framework fundamentally differs from existing approaches in three key aspects. First, unlike existing prompt generation modules and task heads from a single pre-trained model, our PEPT simultaneously leverages multiple pre-trained models to capture diverse aspects of point cloud data. Second, our MLPG module processes features at multiple levels, enabling simultaneous capture of both local geometric details and high-level semantics. Third, our LightHead uses an adaptive fusion mechanism to adjust the models based on the characteristics of the input data and the requirements of the task. This adaptive approach combines the strengths of each model.

3.2. Multi-level prompt generation

Prompt tuning for 3D vision backbones is underexplored, presenting both opportunities and challenges in adapting to the complexity of 3D data. Our work further explores how to develop prompt learning methods specifically designed for 3D data and efficiently utilize pre-trained 3D knowledge.

To this end, we design an MLPG module in each branch of our PEPT framework. The MLPG module generates prompts by processing features across multiple levels, from low-level geometric details to high-level semantic representations. By adopting different MLPG modules to generate prompts for each pre-trained model, this hierarchical approach enables a more comprehensive understanding of point cloud structures while maintaining computational efficiency through minimal trainable parameters.

The MLPG module is illustrated in Fig. 3. A lightweight local neighborhood group convolution (LNGConv) and a multi-level group convolution (MLGConv) are designed for processing point cloud data. In LNGConv, we first construct a kNN graph to identify neighboring points for each sampled center point. Unlike radius-based neighborhood construction, kNN maintains a fixed number of neighboring points for every center point. Consequently, the effective spatial extent of the neighborhood

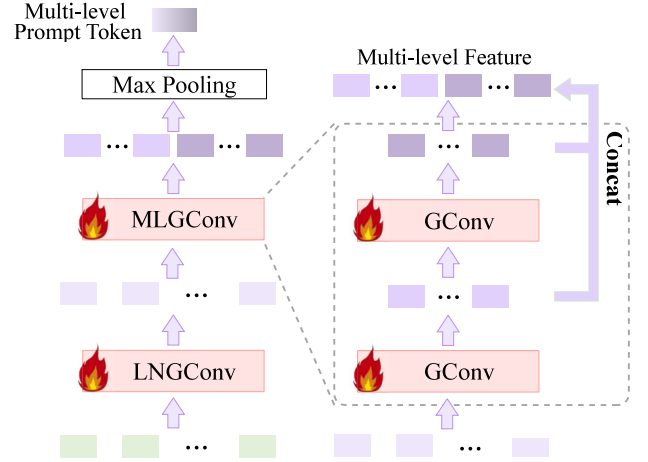


Fig. 3. The architecture of the proposed MLPG module. LNGConv and MLGConv with fewer parameters are used to extract multi-level point cloud features, and the enhanced prompt information is acquired.

varies according to the local point density. In highly sparse regions, some neighbors may be located at relatively larger spatial distances, whereas in dense regions the neighborhood tends to remain more compact.

Then, we apply group convolution (GConv) (Ma et al., 2018; Zhang et al., 2018) to transform these local neighborhood features. The transformed local neighborhood features are aggregated using max-pooling and fed into the MLGConv, which is composed of two GConv layers. Part of the output feature maps is calculated by the first GConv from all input feature maps, and the other part is obtained by the second GConv from the first part. Finally, the multi-level prompt token is obtained after max-pooling on concatenating the first-level feature from the first GConv layer and the second-level feature from the second GConv layer. Although we reduce the overall number of network parameters through group convolution, the output features produced by our MLGConv layer are composed of features from different levels, enabling the module to capture richer representations and thus maintain its performance.

For sparse or non-uniform point cloud regions, the fixed-k strategy may therefore introduce neighboring points that are less representative of the local geometric structure. Nevertheless, this issue can be alleviated in our framework in two aspects. First, the max-pooling operation in LNGConv preserves the most informative local responses while reducing the influence of noisy or less relevant neighboring points. Second, the proposed MLPG module aggregates features from multiple representation levels through MLGConv, and the overall PEPT framework further integrates complementary representations from multiple pre-trained models. Such a multi-level and multi-model design reduces the dependence on a single local neighborhood configuration and helps alleviate the impact of local density variations and non-uniform point distributions.

3.3. Lightweight task head

We describe the proposed LightHead, as shown in Fig. 4. The features extracted by the two branches and the fused features are input into the three classifiers, respectively, to obtain the corresponding prediction results. Given the label matrix Y for supervision, the three sub-loss functions based on Cross Entropy (CE) and Softmax are as follows:

$$Loss_1 = CE(\text{Softmax}(g_1, Y)), \quad (8)$$

$$Loss_2 = CE(\text{Softmax}(g_2, Y)), \quad (9)$$

$$Loss_3 = CE(\text{Softmax}(g_3, Y)). \quad (10)$$

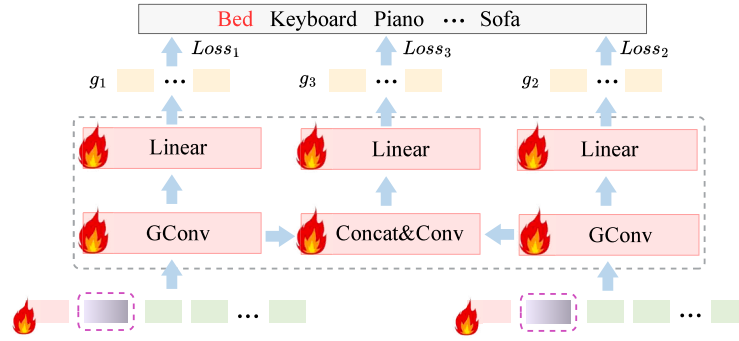


Fig. 4. The architecture of the proposed LightHead. In the training phase, the total loss function is computed as the sum of three sub-losses: $Loss_1$, $Loss_2$, and $Loss_3$. In the test phase, we only calculate the classification results for the fused features g_3 .

Finally, the total loss function is expressed as the sum of the individual sub-losses:

$$Loss = Loss_1 + Loss_2 + Loss_3. \quad (11)$$

Here, we simply sum the prediction losses from all three branches, meaning that their fusion weights are all set to 1. This straightforward fusion mechanism makes our method easy to tune.

4. Experiments

To evaluate the performance of the proposed PEPT framework, we conduct extensive experiments on three benchmark datasets: ScanObjectNN (Uy et al., 2019), ModelNet40 (Wu et al., 2015), and ShapeNetPart (Yi et al., 2016). These experiments aim to demonstrate the effectiveness of PEPT in various 3D point cloud tasks, including object classification, few-shot learning, and part segmentation.

4.1. Object classification

Datasets. We evaluate our method on two object classification datasets: ScanObjectNN (Uy et al., 2019) and ModelNet40 (Wu et al., 2015). ScanObjectNN is a challenging real-world dataset derived from scanned indoor scenes, containing about 15K point cloud samples across 15 categories. The dataset includes occluded objects and cluttered backgrounds, making classification difficult. We conduct experiments on three variants of ScanObjectNN, including OBJ_BG, OBJ_ONLY, and PB_T50_RS. ModelNet40 is a widely used benchmark dataset for 3D object recognition, comprising 12,311 3D CAD models across 40 categories, split into 9843 for training and 2468 for testing. Each point cloud in ModelNet40 is complete, uniform, and noise-free.

Baselines. We compare PEPT with 17 baselines, including PointNet (Qi et al., 2017a), PointNet++ (Qi et al., 2017b), DGCNN (Wang et al., 2019), DTO-Net (Hu et al., 2022), PointNetXt (Qian et al., 2022), PointMLP (Ma et al., 2022), RepSurf-U (Ran et al., 2022), ADS (Hong et al., 2023), OcCo (Wang et al., 2021), Transformer (Yu et al., 2022), Point-BERT (Yu et al., 2022), MaskPoint (Liu et al., 2022b), Point-MAE (Pang et al., 2022), Point-M2AE (Zhang et al., 2022), ACT (Dong et al., 2023), IDPT (Zha et al., 2023), and DAPT (Zhou et al., 2024).

Implementation Details. Following previous work (Ma et al., 2022; Pang et al., 2022; Yu et al., 2022), all experiments are trained for 300 epochs using the SGD optimizer with a weight decay of 0.0002, and the batch size is 32. We set an initial learning rate to 0.02 for PEPT-BM (Point-BERT and Point-MAE as pre-training models) and PEPT-MA (Point-MAE and ACT as pre-training models). All experiments are implemented using the PyTorch (Paszke et al., 2019) framework and conducted on an NVIDIA TITAN Xp GPU. We adopt the Overall Accuracy (OA) to evaluate the performance of our PEPT method. To ensure the reliability and reproducibility of our results, all experiments are randomly repeated 10 times, and we report the best results achieved.

Results. Tables 1 and 2 present the performance comparison on ScanObjectNN and ModelNet40, respectively. We categorize all methods into three distinct groups: supervised learning only, self-supervised learning (full fine-tuning), and self-supervised learning (prompt fine-tuning).

Supervised methods show promising performance on ModelNet40 (89.2% to 95.1%) but struggle with the ScanObjectNN dataset, especially on the more challenging dataset variant PB_T50_RS. Full fine-tuning methods generally outperform supervised approaches, with ACT achieving the highest scores on ScanObjectNN (93.29% on OBJ_BG, 91.91% on OBJ_ONLY, and 88.21% on PB_T50_RS).

Prompt fine-tuning methods adopt Point-BERT (Yu et al., 2022), Point-MAE (Pang et al., 2022), and ACT (Dong et al., 2023) as pre-trained models. On the challenging ScanObjectNN dataset, the proposed PEPT only uses 0.4M trainable parameters, surpassing the full fine-tuning (such as 22.1M for Point-BERT, Point-MAE, and ACT, respectively) and prompt fine-tuning methods (such as 1.7M for IDPT and 1.1M for DAPT). PEPT-BM shows substantial improvements over Point-MAE (+1.89%, +2.24%, +0.25% across the OBJ_BG, OBJ_ONLY, and PB_T50_RS) and even greater gains over Point-BERT (+4.48%, +2.41%, +2.36%). Even on the challenging PB_T50_RS variant, PEPT-BM and PEPT-MA achieve competitive results with minimal trainable parameters. In Tables 1 and 2, the performance gains of our PEPT-BM over the baseline model Point-MAE, and PEPT-MA over the baseline model ACT are presented in brackets. PEPT-BM and PEPT-MA achieve the best or near-best performance in all settings. These results demonstrate that our framework effectively leverages complementary strengths from multiple pre-trained models, achieving more robust and comprehensive point cloud understanding than single-model approaches.

4.2. Few-shot learning

To further evaluate PEPT's effectiveness, we conduct few-shot learning experiments on ModelNet40 using n-way, m-shot configurations ($n \in \{5, 10\}$ and $m \in \{10, 20\}$), consistent with previous works (Pang et al., 2022; Yu et al., 2022; Zha et al., 2023; Zhou et al., 2024). The results shown in Table 3 suggest that our PEPT-BM achieved the best performance across all settings with only 0.4M trainable parameters, outperforming both full fine-tuning (Point BERT and Point MAE) and prompt tuning methods (IDPT and DAPT), which demonstrates the effectiveness of transferring multiple pre-trained models' knowledge to data-scarce scenarios.

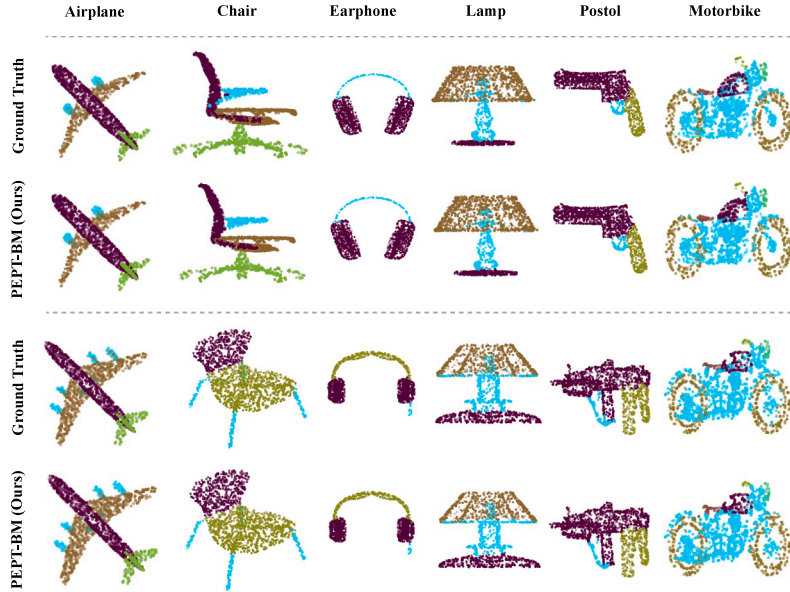
4.3. Part segmentation

Dataset. ShapeNetPart (Yi et al., 2016) is a 3D part segmentation dataset. It consists of 16,881 samples from 16 shape categories. We follow previous work (Zha et al., 2023), and introduce prompts into the inputs of the 3rd, 7th, and 11th layers.

Table 1

Performance comparison on ScanObjectNN. ‘TP’ denotes trainable parameters.

Methods	TP (M)	OBJ_BG	OBJ_ONLY	PB_T50_RS
Supervised Learning Only				
PointNet (Qi et al., 2017a)	3.5	73.3	79.2	68.0
PointNet++ (Qi et al., 2017b)	1.5	82.3	84.3	77.9
DGCNN (Wang et al., 2019)	1.8	82.8	86.2	78.1
DTO-Net (Hu et al., 2022)	–	85.5	86.9	80.8
PointNeXt (Qian et al., 2022)	1.4	–	–	87.7
PointMLP (Ma et al., 2022)	13.2	–	–	85.4
RepSurf-U (Ran et al., 2022)	1.5	–	–	84.3
ADS (Hong et al., 2023)	–	–	–	87.5
Self-Supervised Learning (Full fine-tuning)				
OcCo (Wang et al., 2021)	22.1	84.85	85.54	78.79
Transformer (Yu et al., 2022)	22.1	79.86	80.55	77.24
Point-BERT (Yu et al., 2022)	22.1	87.43	88.12	83.07
MaskPoint (Liu et al., 2022b)	22.1	89.70	89.30	84.60
Point-MAE (Pang et al., 2022)	22.1	90.02	88.29	85.18
Point-M2AE (Zhang et al., 2022)	15.3	91.22	88.81	86.43
ACT (Dong et al., 2023)	22.1	93.29	91.91	88.21
Self-Supervised Learning (Prompt tuning)				
Point-MAE (Pang et al., 2022) (Pre-trained model)	22.1	90.02	88.29	85.18
+ IDPT (Zha et al., 2023)	1.7	91.22 (+1.20)	90.02 (+1.73)	84.94 (−0.24)
+ DAPT (Zhou et al., 2024)	1.1	90.88 (+0.86)	90.19 (+1.90)	85.08 (−0.10)
PEPT-BM (Ours)	0.4	91.91 (+1.89)	90.53 (+2.24)	85.43 (+0.25)
ACT (Dong et al., 2023) (Pre-trained model)	22.1	93.29	91.91	88.21
+ IDPT (Zha et al., 2023)	1.7	93.12 (−0.17)	92.26 (+0.35)	87.65 (−0.56)
PEPT-MA (Ours)	0.4	93.80 (+0.51)	92.77 (+0.86)	88.45 (+0.24)

**Fig. 5.** Parts segmentation visualization on ShapeNetPart.

Baselines. We compare PEPT with 9 baselines, including PointNet (Qi et al., 2017a), PointNet++ (Qi et al., 2017b), DGCNN (Wang et al., 2019), OcCo (Wang et al., 2021), Point-BERT (Yu et al., 2022), Point-MAE (Pang et al., 2022), ACT (Dong et al., 2023), IDPT (Zha et al., 2023) and DAPT (Zhou et al., 2024).

Results. Table 4 presents the quantitative results of the part segmentation on the ShapeNetPart dataset. The table employs two metrics $mIoU_c$ (%) and $mIoU_f$ (%) for the assessment. Fig. 5 shows the segmentation result intuitively. As we can see, the segmentation results of our PEPT-BM are very close to the ground-truth segmentation and can accurately discern the different parts of various objects. Our PEPT-BM, which leverages Point-BERT and Point-MAE as pre-training models, achieves competitive performance compared with the full fine-tuning method Point-MAE, with much fewer trainable parameters.

4.4. Scene segmentation

Dataset. The Stanford Large-Scale 3D Indoor Spaces dataset (S3DIS) (Armeni et al., 2016) comprises 3D point clouds from 271 rooms across 6 large indoor areas. Semantic annotations are provided at the point level, covering 13 semantic categories (e.g., ceiling, floor, wall, table, chair). Consistent with prior work (Tchapmi et al., 2017), we use Area 5 as the test split for evaluation.

Baselines. We compare PEPT with 8 baselines, including PointNet (Qi et al., 2017a), PointNet++ (Qi et al., 2017b), PointCNN (Li et al., 2018), PCT (Guo et al., 2021), Point-MAE (Pang et al., 2022), ACT (Dong et al., 2023), IDPT (Zha et al., 2023) and DAPT (Zhou et al., 2024).

Results. Table 5 presents the quantitative results of the scene segmentation on the S3DIS dataset. The table employs two metrics $mAcc$

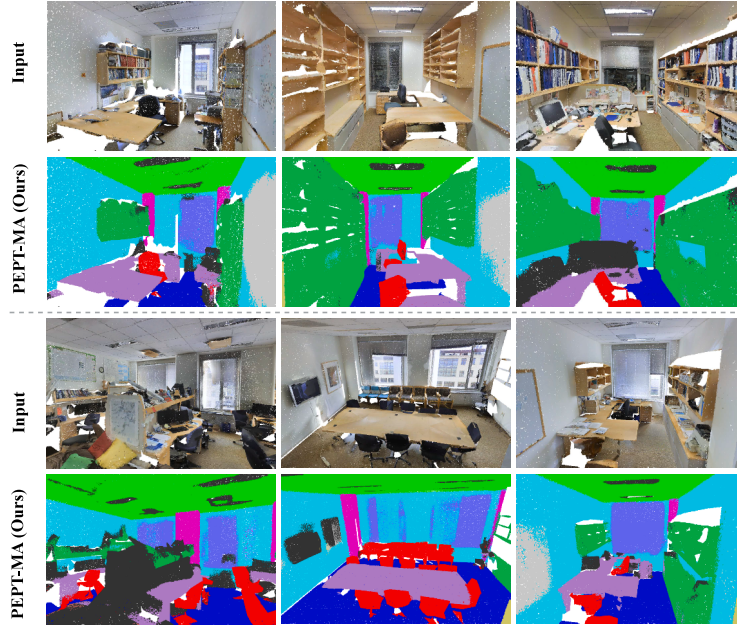


Fig. 6. Scene segmentation visualization on S3DIS.

Table 2

Performance comparison on ModelNet40. ‘pnt’ denote points.

Methods	TP (M)	Inputs	OA (%)
Supervised Learning Only			
PointNet (Qi et al., 2017a)	3.5	1K pnt	89.2
PointNet++ (Qi et al., 2017b)	1.5	1K pnt	90.7
DGCNN (Wang et al., 2019)	1.8	1K pnt	92.9
DTO-Net (Hu et al., 2022)	–	1K pnt	93.3
PointNeXt (Qian et al., 2022)	1.4	1K pnt	94.0
PointMLP (Ma et al., 2022)	13.2	1K pnt	94.5
RepSurf-U (Ran et al., 2022)	1.5	1K pnt	94.4
ADS (Hong et al., 2023)	–	1K pnt	95.1
Self-Supervised Learning (Full fine-tuning)			
OcCo (Wang et al., 2021)	22.1	1K pnt	92.1
Transformer (Yu et al., 2022)	22.1	1K pnt	91.4
Point-BERT (Yu et al., 2022)	22.1	1K pnt	93.2
MaskPoint (Liu et al., 2022b)	22.1	1K pnt	93.8
Point-MAE (Pang et al., 2022)	22.1	1K pnt	93.8
Point-M2AE (Zhang et al., 2022)	15.3	1K pnt	94.0
ACT (Dong et al., 2023)	22.1	1K pnt	93.7
Self-Supervised Learning (Prompt tuning)			
Point-MAE (Pang et al., 2022) (Pre-trained model)	22.1	1K pnt	93.8
+ IDPT (Zha et al., 2023)	1.7	1K pnt	94.4 (+0.6)
+ DAPT (Zhou et al., 2024)	1.1	1K pnt	94.0 (+0.2)
PEPT-BM (Ours)	0.4	1K pnt	93.9 (+0.1)

(%) and mIoU (%) for the assessment of this dataset. As we can see in the table, compared with prompt tuning methods (IDPT and DAPT), our PEPT-MA achieves the best performance on this task. In Fig. 6, we have also provided several visualized segmentation results to show the effectiveness of our method more intuitively.

4.5. Ablation study

In this subsection, to demonstrate the effectiveness of our PEPT framework, we systematically evaluate the key components and hyper-parameters of our framework through comprehensive ablation studies on the ScanObjectNN (OBJ_BG) dataset. To validate our design choices and provide the optimal configuration for our model, our investigations span comparisons of single-model and multi-model approaches,

feature fusion manner, loss function terms, group numbers, network depth, prompt insertion position, and learning rate.

Single-model vs. Multi-model Prompt Tuning. We first evaluate the effectiveness of our multi-model prompt tuning approach compared to single-model prompt tuning. Table 6 presents the comparison results of using Point-BERT or Point-MAE individually as pre-trained networks, versus our proposed multi-model approach PEPT-BM, which utilizes both. The findings demonstrate a significant improvement in OA when employing multi-model prompt tuning. Specifically, we can observe that the performance of our multi-model strategy increases from 89.67% (with Point-Bert) and 91.05% (with Point-MAE) to 91.91%, achieving significant improvements of 2.24% and 0.86%, respectively.

Besides, we can note that our PEPT-BERT and PEPT-MAE in Table 6 have higher performance as well as many fewer trainable parameters than Point-BERT and Point-MAE. Specifically, the baseline methods Point-BERT (Yu et al., 2022) and Point-MAE (Pang et al., 2022) require full fine-tuning of the models, i.e., the entire networks need to be fine-tuned, resulting in 22.1M trainable parameters. In contrast, in our method, each branch (PEPT-BERT or PEPT-MAE) only fine-tunes the task head and the MLPG module, with a total of 0.1M parameters. Consequently, the number of trainable parameters in LNGConv and MLGConv for each branch is less than 0.1M. For the dual-branch PEPT-BM, the trainable parameters in LNGConv and MLGConv are less than 0.2M, which shows a significant reduction compared to the 22.1M parameters required by the baseline methods.

Feature Fusion Manner. We explore the impact of different fusion manners of the features from different pre-trained branches. Specifically, we compare two common fusion operations: concatenation and addition. As shown in Table 7, the concatenation operation outperforms the addition operation, with OA increasing from 91.22% to 91.91%. This improvement indicates that the concatenation operation is more effective in integrating complementary features from the two pre-trained models, allowing the network to capture richer point cloud characteristics.

Contribution of Different Loss Terms. Our loss function, as defined in Eq. (11), is composed of three sub-loss terms: $Loss_1$, $Loss_2$, and $Loss_3$. $Loss_1$ corresponds to the loss of the Point-Bert branch, $Loss_2$ corresponds to the loss of the Point-MAE branch, and $Loss_3$ corresponds to the loss after feature fusion. To validate the impacts of these loss terms, we have conducted an ablation study for various combinations of these

Table 3
Comparison of few-shot learning performance on ModelNet40.

Methods	5-way		10-way	
	10-shot	20-shot	10-shot	20-shot
Self-Supervised Learning (Full fine-tuning)				
OcCo (Wang et al., 2021)	94.0±3.6	95.9±2.3	89.4±5.1	92.4±4.6
Point-BERT (Yu et al., 2022)	94.6±3.1	96.3±2.7	91.0±5.4	92.7±5.1
Point-MAE (Pang et al., 2022)	96.3±2.5	97.8±1.8	92.6±4.1	95.0±3.0
Point-M2AE (Zhang et al., 2022)	96.8±1.8	98.3±1.4	92.3±4.5	95.0±3.0
ACT (Dong et al., 2023)	96.8±2.3	98.0±1.4	93.3±4.0	95.6±2.8
Self-Supervised Learning (Prompt tuning)				
Point-BERT (Yu et al., 2022) (Pre-trained model)	94.6±3.1	96.3±2.7	91.0±5.4	92.7±5.1
+ IDPT (Zha et al., 2023)	96.0±1.7	97.2±2.6	91.9±4.4	93.6±3.5
+ DAPT (Zhou et al., 2024)	95.8±2.1	97.3±1.3	92.2±4.3	94.2±3.4
Point-MAE (Pang et al., 2022) (Pre-trained model)	96.3±2.5	97.8±1.8	92.6±4.1	95.0±3.0
+ IDPT (Zha et al., 2023)	97.3±2.1	97.9±1.1	92.8±4.1	95.4±2.9
+ DAPT (Zhou et al., 2024)	96.8±1.8	98.0±1.0	93.0±3.5	95.5±3.2
PEPT-BM (Ours)	97.4±1.6	98.8±1.1	93.4±3.8	95.9±2.9

Table 4
Comparison of experimental results on ShapeNetPart.

Methods	TP (M)	mIoU _c	mIoU _f
Supervised Learning Only			
PointNet (Qi et al., 2017a)	–	80.39	83.7
PointNet++ (Qi et al., 2017b)	–	81.85	85.1
DGCNN (Wang et al., 2019)	–	82.33	85.2
Self-Supervised Learning (Full fine-tuning)			
OcCo (Wang et al., 2021)	27.09	83.42	85.1
Point-BERT (Yu et al., 2022)	27.09	84.11	85.6
Point-MAE (Pang et al., 2022)	27.06	84.19	86.1
ACT (Dong et al., 2023)	27.06	84.66	86.1
Self-Supervised Learning (Prompt tuning)			
Point-MAE (Pang et al., 2022) (Pre-trained model)	27.06	84.19	86.1
+ IDPT (Zha et al., 2023)	5.69	83.79	85.7
+ DAPT (Zhou et al., 2024)	5.65	84.01	85.7
PEPT-BM (Ours)	5.09	83.74	85.8

Table 5
Comparison of experimental results on S3DIS.

Methods	mAcc (%)	mIoU (%)
Supervised Learning Only		
PointNet (Qi et al., 2017a)	49.0	41.1
PointNet++ (Qi et al., 2017b)	67.1	53.5
PointCNN (Li et al., 2018)	63.9	57.3
PCT (Guo et al., 2021)	67.7	61.3
Self-Supervised Learning (Full fine-tuning)		
Point-MAE (Pang et al., 2022)	69.9	60.8
ACT (Dong et al., 2023)	71.1	61.2
Self-Supervised Learning (Prompt tuning)		
ACT (Dong et al., 2023) (Pre-trained model)	71.1	61.2
+ IDPT (Zha et al., 2023)	65.0	53.8
+ DAPT (Zhou et al., 2024)	65.6	55.1
PEPT-MA (Ours)	67.4	58.0

sub-losses. The experimental results in Table 8 indicate that the PEPT framework achieves optimal performance when all three sub-loss terms are employed simultaneously. The removal of any single term results in performance degradation. This observation not only substantiates the efficacy of our loss function design but also highlights the importance of learning comprehensive feature representations from multiple models. Each loss term contributes unique and complementary information,

Table 6
The impact of the single-model and multi-model prompt tuning on the model performance.

Method	Pre-trained Network		TP (M)	OA (%)
	Point-BERT	Point-MAE		
Self-Supervised Learning (Full fine-tuning)				
Point-BERT (Yu et al., 2022)	✓		22.1	87.43
Point-MAE (Pang et al., 2022)		✓		90.02
Self-Supervised Learning (Prompt tuning)				
PEPT-BERT (Ours)	✓		0.1	89.67
PEPT-MAE (Ours)		✓		91.05
PEPT-BM (Ours)	✓	✓	0.4	91.91

Table 7
The impact of the fusion manner on the model performance.

Method	Fusion manner	TP (M)	OA (%)
PEPT-BM	Add.	0.3	91.22
	Concat.	0.4	91.91

Table 8
The impact of the different loss terms on the model performance.

Method	Loss ₁	Loss ₂	Loss ₃	TP (M)	OA (%)
PEPT-BM		✓	✓	0.4	90.53
	✓		✓		91.22
	✓	✓	✓		91.91

Table 9
The impact of the different number of groups on the model performance.

Method	Group	TP (M)	OA(%)
PEPT-BM	2	0.62	91.57
	8	0.4	91.91
	32	0.36	91.39
	64	0.34	91.05

enhancing the model's ability to recognize complex point cloud structures.

Impact of Group Numbers. Since we use LNGConv in our MLGP network, we explore the impacts of the number of groups of LNGConv on the model performance. As detailed in Table 9, we find that increasing the number of groups has a small impact on the number of trainable

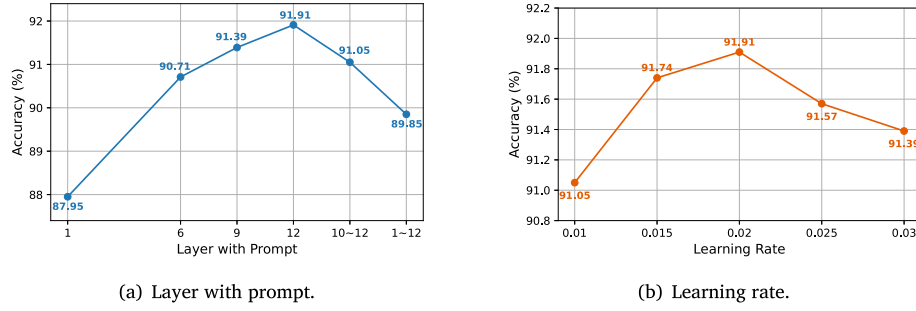


Fig. 7. The impact of the position of prompt insertion and the learning rate on the performance.

Table 10

The impact of the different number of LNGConv layers on the model performance.

Methods	TP (M)	OA (%)
1 LNGConv	0.4	91.91
2 LNGConv	0.56	91.91
3 LNGConv	0.7	91.39

Table 11

The impact of the different convolution kernel sizes on the model performance.

Method	Kernel size	TP (M)	OA(%)
PEPT-BM	1	0.4	91.91
	3	1.22	90.53

parameters. However, too many groups result in a bit of network performance degradation. Taking both parameter efficiency and accuracy into consideration, we set the number of groups to 8. This configuration achieves satisfactory performance while maintaining computational efficiency, representing an optimal trade-off between training cost and inference accuracy. Similarly, by exploring the number of groups for two GConvs in the MLGConv module, both were set to 2, and the number of groups of two GConvs in the task head is 384.

Depth of LNGConv Layers. We explore the impact of its number of trainable layers on the experimental results. As shown in Table 10, as the number of LNGConv layers increases, the number of parameters increases, but it has not improved the network performance. Based on these observations, we conclude that a single LNGConv layer achieves the optimal balance between model performance and parameters.

Convolution Kernel Size. Table 11 presents the impact of convolution kernel size in LNGConv and MLGConv on network performance, comparing the experimental results of convolution operations with kernel sizes of 1 and 3. As can be observed from the table, the performance achieved by kernel size 1 is higher. Consequently, we set the convolution kernel to 1.

Position of Prompt Insertion. We evaluate the impact of inserting our prompt into different pre-trained model layers. As illustrated in Fig. 7(b), as the insertion position progresses from the 1st layer to the 12th layer, the accuracy gradually increases from 87.95% to a peak of 91.91%. The optimal performance is achieved when only the prompt is applied to layer 12. It suggests that deeper layers, which contain more semantically rich representations of the point cloud data, better leverage our prompt mechanism. Consequently, we insert the prompt solely into the final layer 12, which balances performance and computational efficiency. Besides, when applying the shared prompts to multiple layers simultaneously, such as layers 10~12 (91.05%) or all layers 1~12 (89.85%), the performance becomes worse. This phenomenon can be attributed to the interference between prompts at different hierarchical

levels of feature representation, which is consistent with the finding of IDPT (Zha et al., 2023).

Learning Rate Analysis. Fig. 7(b) illustrates the relationship between the learning rate and the accuracy of the model at five distinct learning rates: 0.01, 0.015, 0.02, 0.025, and 0.03. We observe that accuracy increases as the learning rate increases, reaching a peak value of 91.91% at a learning rate of 0.02. Further increasing the learning rate to 0.025 and 0.03 leads to a decrease in accuracy, dropping to 91.57% and 91.39%, respectively. The experimental results suggest that 0.02 represents the optimal learning rate for our experiment.

Different combinations of pre-trained models. Here, we further explore more combinations of pre-trained models. In addition to the previously investigated PEPT-BM and PEPT-MA, we introduce two new combinations: PEPT-MG, which uses Point-MAE (Pang et al., 2022) and PointGPT-S (Chen et al., 2023) as the pre-trained models, and PEPT-GA, which uses PointGPT-S (Chen et al., 2023) and ACT (Dong et al., 2023) as the pre-trained models. We apply these combinations to the object classification task on the ScanObjectNN dataset, and the final results are reported in Table 12. For each combination, we also present the absolute performance gains over the full fine-tuning results of the best-performing single pre-trained model. Although PEPT-MG and PEPT-GA achieved consistent improvements on the OBJ_BG and OBJ_ONLY variants of ScanObjectNN, their performance gains on the more challenging PB_T50_RS benchmark were relatively limited. One possible explanation is that PB_T50_RS contains stronger geometric perturbations, including background clutter, random translations, and rotations, which place greater demands on the robustness and complementarity of the learned representations. Under such challenging conditions, different pre-trained models may capture overlapping geometric patterns and become more susceptible to similar perturbation-induced errors. As a result, the complementary information available for feature fusion may be reduced, limiting the effectiveness of combining multiple pre-trained models. These observations suggest that the performance of the proposed framework depends not only on the number of incorporated pre-trained models but also on the diversity and complementarity of the representations they learn. In Table 13, we additionally report the mean performance and standard deviation (mean±std) over 10 random runs for different model combinations. It can be observed that our methods are generally robust and reliable.

4.6. Efficiency study

To further verify the efficiency of our method, we compare the training and inference speeds of full fine-tuning and prompt-tuning approaches. Specifically, we measure the throughput in terms of the number of processed samples per second for each method. The final results are reported in Table 14. It can be observed that our PEPT-BERT achieves outstanding training and inference speeds of 355 and 726 samples per second, respectively, representing improvements of 38.1% and 25.0% over the full fine-tuning Point-BERT. This is mainly attributed to

Table 12
Object classification performance of different combinations of pre-trained models.

Methods	TP (M)	OBJ_BG	OBJ_ONLY	PB_T50_RS
Point-MAE (Pang et al., 2022) (Pre-trained model)	22.1	90.02	88.29	85.18
PEPT-BM (Ours)	0.4	91.91 (+1.89)	90.53 (+2.24)	85.43 (+0.25)
ACT (Dong et al., 2023) (Pre-trained model)	22.1	93.29	91.91	88.21
PEPT-MA (Ours)	0.4	93.80 (+0.51)	92.77 (+0.86)	88.45 (+0.24)
PointGPT-S (Chen et al., 2023) (Pre-trained model)	29.2	91.60	90.00	86.90
PEPT-MG (Ours)	0.4	92.77 (+1.17)	91.05 (+1.05)	86.43 (−0.47)
ACT (Dong et al., 2023) (Pre-trained model)	22.1	93.29	91.91	88.21
PEPT-GA (Ours)	0.4	93.29 (+0.00)	92.08 (+0.17)	88.10(−0.11)

Table 13
Object classification performance (mean±std) of 10 random runs of different combinations of pre-trained models.

Methods	OBJ_BG	OBJ_ONLY	PB_T50_RS
PEPT-BM (Ours)	90.95±0.53	89.69±0.51	84.88±0.39
PEPT-MA (Ours)	93.10±0.36	92.08±0.38	87.95±0.37
PEPT-MG (Ours)	91.95±0.56	90.40±0.52	85.68±0.50
PEPT-GA (Ours)	92.58±0.38	91.36±0.32	87.87±0.24

our lightweight prompt generation module and lightweight task head, which substantially reduce computational overhead while preserving the model’s representation capability. Notably, for Point-MAE, the training and inference speeds slightly decrease after applying IDPT and DAPT prompting methods. In contrast, our PEPT-MAE is faster than both Point-MAE + IDPT and Point-MAE + DAPT in both training and inference speeds, owing to the lightweight design of our task head. Moreover, although our PEPT-BM consists of two pre-trained branches, it is still faster than the prompt-based methods Point-MAE + IDPT and Point-MAE + DAPT, further demonstrating the high efficiency of our approach.

In addition to throughput, we further compare the peak GPU memory utilization of different fine-tuning strategies, as reported in Table 14. PEPT-BERT requires only 4.4 GB of peak GPU memory while using merely 0.1M trainable parameters, which is substantially lower than the 9.3 GB required by full fine-tuning Point-BERT. For PEPT-

MAE and PEPT-BM, the peak GPU memory utilization is higher than that of some prompt-tuning baselines. This difference is largely attributed to the local neighborhood context modeling strategy. Specifically, PEPT-MAE, which is built upon the Point-MAE architecture, generates prompts from the neighborhoods of 128 sampled points, whereas PEPT-BERT, based on Point-BERT, processes only 64 sampled points. PEPT-MAE operates on a larger number of local neighborhoods, leading to higher peak GPU memory utilization. Similarly, PEPT-BM inherits the neighborhood processing mechanism from both Point-BERT and Point-MAE branches, and the dual-branch architecture further introduces feature fusion operations, resulting in a slightly higher peak memory consumption.

Nevertheless, the increased memory overhead enables richer local geometric context modeling and contributes to improved classification performance. Despite the increased peak GPU memory utilization, PEPT-MAE and PEPT-BM achieve superior classification performance while maintaining a very small number of trainable parameters and competitive throughput. In particular, PEPT-BM achieves the best overall accuracy of 91.91% with only 0.4M trainable parameters.

Overall, the proposed PEPT framework achieves a favorable trade-off among trainable parameters, GPU memory utilization, throughput, and task performance, demonstrating its effectiveness for parameter-efficient adaptation of pre-trained point cloud models.

Table 14
Peak GPU memory, training and inference speed (samples/second) were measured on a single NVIDIA TITAN Xp GPU.

Method	TP (M)	Memory (GB)	Training Speed	Inference Speed	OA (%)
Self-Supervised Learning (Full fine-tuning)					
Point-BERT	22.1	9.3	257	581	87.42
Point-MAE		4.6	42	53	90.02
Self-Supervised Learning (Prompt tuning)					
Point-MAE + IDPT	1.7	4.4	36	48	91.22
Point-MAE + DAPT	1.1	2.2	37	53	90.88
PEPT-BERT (Ours)	0.1	4.4	355	726	89.67
PEPT-MAE (Ours)	0.1	7.4	46	58	91.05
PEPT-BM (Ours)	0.4	7.5	41	52	91.91

5. Conclusion and limitation discussion

In this paper, we introduce PEPT, a novel framework for 3D point cloud understanding. By introducing multiple pre-trained point cloud models and proposing MLPG modules, our method significantly enhances feature extraction ability. Extensive experiments on the ModelNet40, ScanObjectNN, and ShapeNetPart datasets demonstrate that PEPT achieves competitive or superior performance compared to existing methods while significantly reducing trainable parameters.

Our method also has several limitations. First, although it supports multiple branches of pre-trained models, our experiments show that using two branches already achieves strong performance. In practice, adding more branches may bring further improvements, but the marginal gain is likely to diminish as the number of branches increases. The optimal number of pre-trained branches, therefore, needs to be determined empirically based on the specific task and dataset at hand. Second, in the lightweight task head, the losses from each branch are summed with equal weights. Although this simple and efficient strategy has achieved competitive performance, a dynamic weighting scheme may better optimize the contribution of each pre-trained model during the learning process. Exploring such dynamic weighting strategies remains an important direction for future work. Third, our model requires access to fully visible training data. In decentralized scenarios (Yang et al., 2025), however, full data collection is often impossible due to privacy constraints. Extending our method to the federated learning paradigm may thus represent a promising direction for future research.

CRedit authorship contribution statement

Xinyan Huang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Fang Liu:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition; **Licheng Jiao:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition; **Yaming Yang:** Writing – review & editing, Validation, Methodology, Funding acquisition; **Shuo Li:** Writing – review & editing, Validation, Methodology, Investigation; **Puhua Chen:** Writing – review & editing, Methodology; **Lingling Li:** Writing – review & editing, Formal analysis; **Xu Liu:** Writing – review & editing, Validation; **Wenping Ma:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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